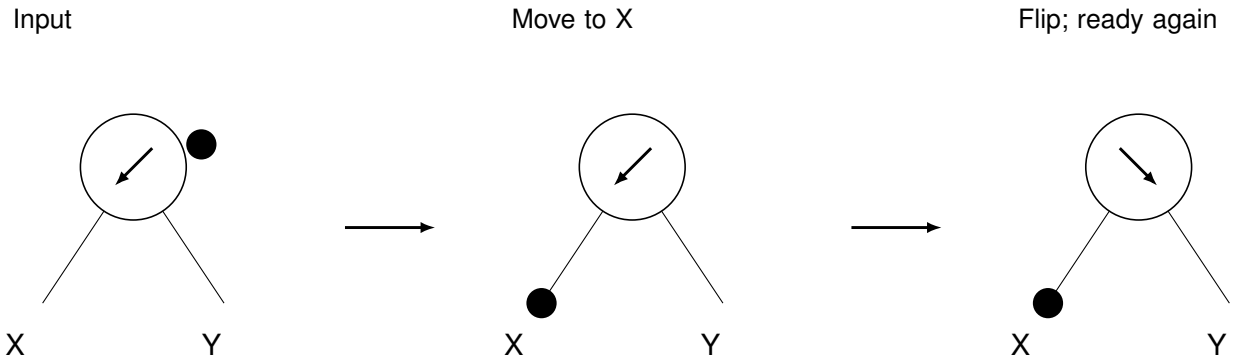
Send counters to the exits, one at a time. At each circle, follow the NEXT arrow, then flip its card. Leave every card as it is for the next counter. Keep the counter beside the card so its arrow stays visible. A partner checks each move.



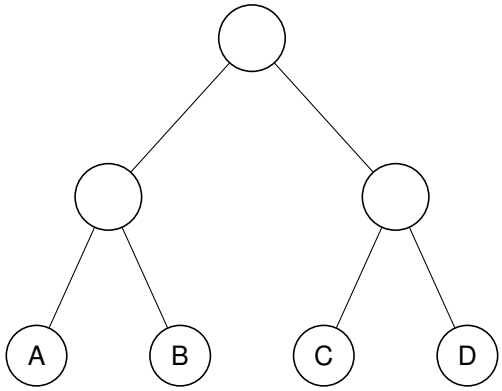
Problem 1: Use the one-switch board. Choose its first arrow. Send 12 counters through and keep their arrival order. Then start a new trial with the other first arrow.

Arrival order

Arrival order

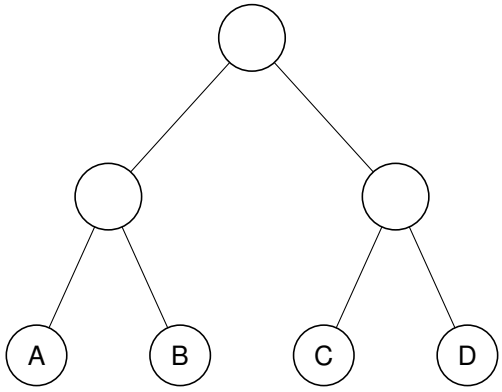
Which arrival strings can this switch make: XYXY, XXYY, YXYX, XYYX?

Problem 2: Use the four-exit tree. Choose starting arrows that send the first counter to D. Keep those cards and send seven more counters. Record the starting arrows and arrival order.



Arrival order

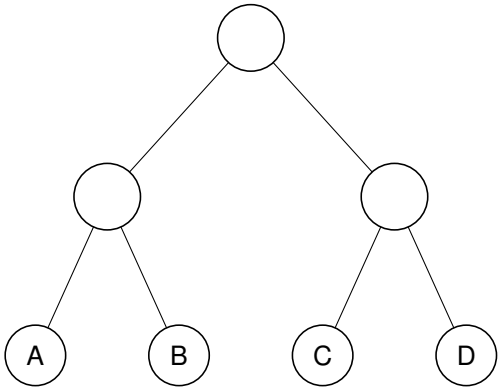
What can you change in the starting arrows while keeping D first? Find every arrival order.



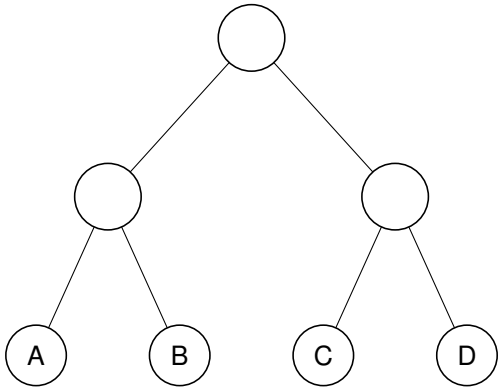
Arrival order

Problem 3: Program the four-exit tree to make each arrival order below. Start a new trial for each order. If an order is impossible, explain what stops it.

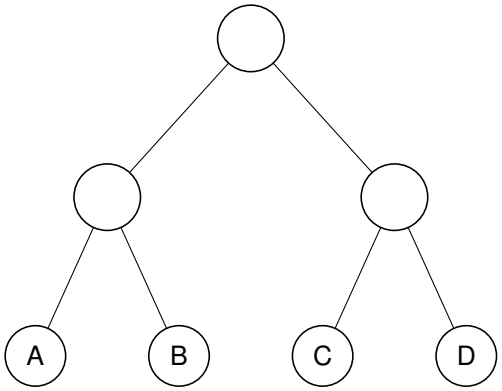
A C B D



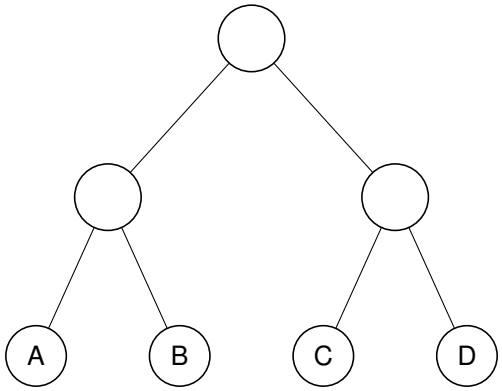
B D A C



A B C D

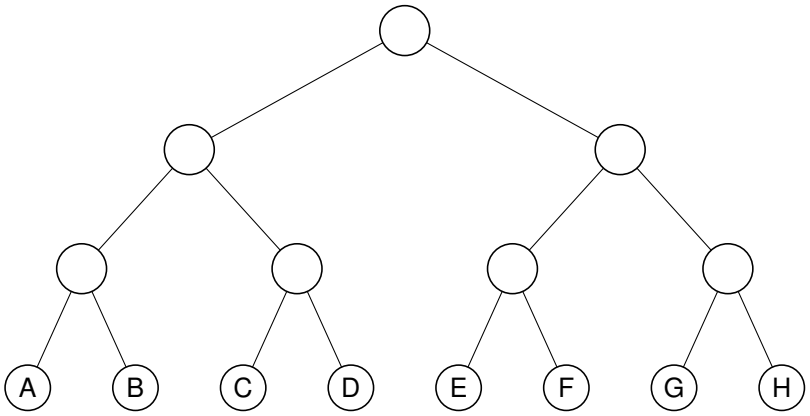


D A B C

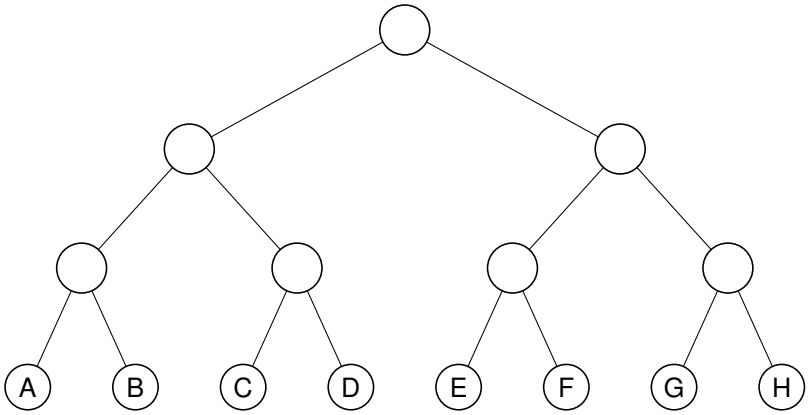


Problem 4: Use the eight-exit tree. Choose starting arrows to make each order. Which orders can the machine make?

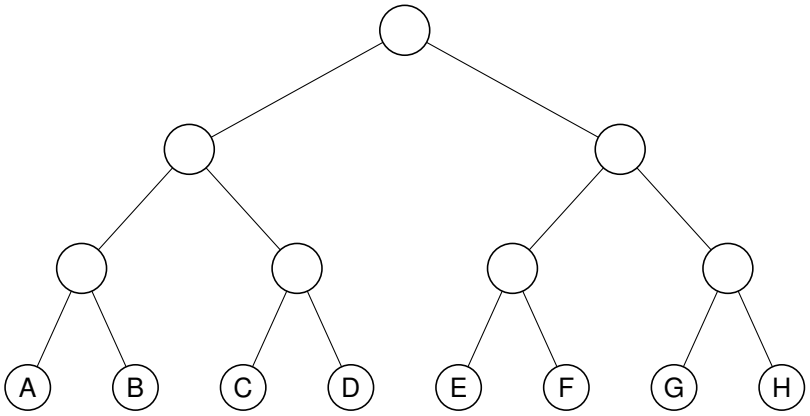
A E C G B F D H



H D F B G C E A



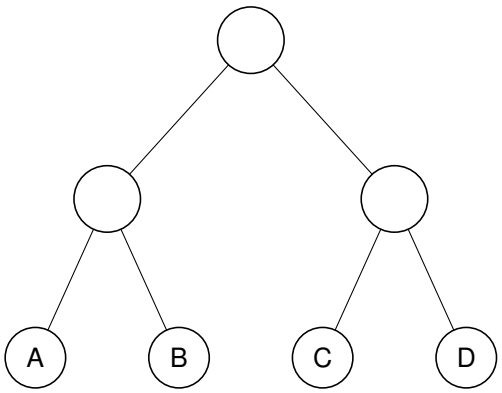
A E B F C G D H

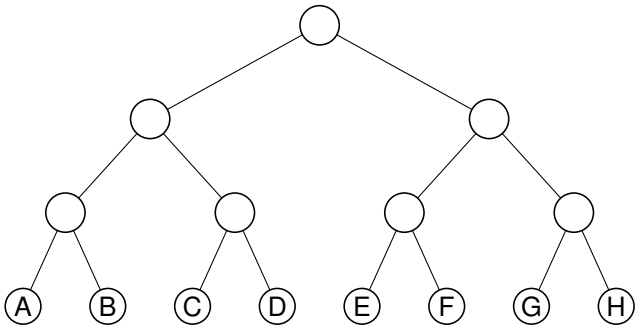


Problem 5: Invent an eight-counter order that visits every exit once but cannot be made by the eight-exit tree. Explain why it cannot work.

Arrival order

Problem 6: Choose any starting arrows on either tree. Send four counters through the four-exit tree or eight through the eight-exit tree. Must every exit get a counter before any exit gets a second one? Must the arrows return to their starting positions? Explain what makes your answers work for every starting position.





Problem 7: Make a second way to choose exits: toss a coin for each counter, sending heads to X and tails to Y. Compare 12 coin choices with 12 arrivals from the one-switch board.

Coin

Switch

Find a six-counter arrival order that the coin could make but the switch could never make. Could the coin make every order that the switch can make?